

HW7 (Chapter 18) for MSE 201:002 Spring 2023

Textbook:

W. D. Callister and D.G. Rethwisch, Materials Science and Engineering: An Introduction, 8th Edition

Chapter 18

- 18.14 (a)** Using the data in Figure 18.8, determine the values of ρ_0 and a from Equation 18.10 for pure copper. Take the temperature T to be in degrees Celsius. **(b)** Determine the value of A in Equation 18.11 for nickel as an impurity in copper, using the data in Figure 18.8. **(c)** Using the results of parts (a) and (b), estimate the electrical resistivity of copper containing 1.75 at% Ni at 100°C.
- 18.17** A cylindrical metal wire 2 mm (0.08 in.) in diameter is required to carry a current of 10 A with a minimum of 0.03 V drop per foot (300 mm) of wire. Which of the metals and alloys listed in Table 18.1 are possible candidates?
- 18.30** Germanium to which $5 \times 10^{22} \text{ m}^{-3}$ Sb atoms have been added is an extrinsic semiconductor at room temperature, and virtually all the Sb atoms may be thought of as being ionized (i.e., one charge carrier exists for each Sb atom). **(a)** Is this material n -type or p -type? **(b)** Calculate the electrical conductivity of this material, assuming electron and hole mobilities of 0.1 and 0.05 $\text{m}^2/\text{V}\cdot\text{s}$, respectively.
- 18.40** Estimate the electrical conductivity, at 85°C, of silicon that has been doped with 10^{20} m^{-3} of phosphorus atoms.

- 18.D4** Specify an acceptor impurity type and concentration (in weight percent) that will produce a *p*-type silicon material having a room temperature electrical conductivity of $50\ (\Omega\cdot\text{m})^{-1}$.